

JUL-SINTECE Integration Control Document



CNCA Certificate Issuance Process

Amendment and Cancellation Capabilities

Version 2.0
Abu Dhabi Ports
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Document Control

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Version	Date	Author	Description of Changes
1.0	2025-11-12	Abu Dhabi Ports	Initial ICD creation with technical specifications for JUL-SINTECE integration. Defined API specifications, data models, integration workflows, security requirements, and testing procedures.
2.0	2025-11-13	Abu Dhabi Ports	Added amendment and cancellation APIs, approval workflows, status management with canAmend/canCancel flags, validation framework, and error handling. Added process flow improvements and implementation recommendations.



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1. Introduction

1.1 Purpose

This Interface Control Document (ICD) defines the technical specifications and interface requirements for integrating the JUL system (Abu Dhabi Ports) with the SINTECE system (ARCCLA) for CNCA certificate issuance.

Key Features:

- Certificate submission and approval workflow
- Amendment and cancellation requests
- Master data synchronization
- Status tracking and notifications

1.2 Scope

In Scope:

- Certificate request lifecycle (creation to issuance)
- Amendment and cancellation workflows
- API specifications and data models
- Authentication and authorization
- Validation and error handling

Out of Scope:

- Internal SINTECE business logic
- Payment gateway implementation
- Mobile application development
- Legacy system migration

1.3 Audience

- **Developers:** API implementation and integration
- **Architects:** System design and technical decisions
- **QA Engineers:** Testing and validation
- **Business Analysts:** Requirements validation
- **Project Managers:** Implementation tracking

1.4 Key Terms

Term	Definition
ARCCLA	Angolan regulatory agency responsible for cargo certification
BL	Bill of Lading - shipping document required for certificates
CNCA	Certificado Nacional de Carga de Angola - required certificate for Angola cargo
CTN	Certificate/Cargo Tracking Note - internal certificate reference
DUP	Declaration of Unique Property - unique customs identifier
JUL	Abu Dhabi Ports system for CNCA certificate management

SINTECE	ARCCLA system for certificate processing and approval
Amendment	Request to modify submitted certificate data
Cancellation	Request to cancel submitted certificate
canAmend/canCancel	System flags indicating eligibility for amendment/cancellation

2. System Overview

2.1 System Architecture

JUL System (Abu Dhabi Ports)

- Trader portal for CNCA certificate management
- User interface for request creation and document upload
- Amendment/cancellation request capabilities
- Payment processing and certificate download
- Real-time status tracking and notifications

SINTECE System (ARCCLA)

- Backend certificate processing and approval system
- ARCCLA broker review workflow
- Amendment/cancellation approval processes
- Invoice generation and certificate issuance
- Master data management and reporting

Integration Layer

- RESTful APIs with JSON data format
- JWT authentication and secure communication
- Real-time API calls and status notifications
- Error handling and audit logging

2.2 Integration Pattern

Communication:

- RESTful APIs with standard HTTP methods (GET, POST, PUT, DELETE)
- JSON data format with schema validation
- HTTPS/TLS encryption and JWT authentication
- OData query support for filtering and pagination

Data Flow:

1. JUL validates certificate data and submits request
2. SINTECE processes with business rules validation
3. ARCCLA broker reviews and approves/rejects
4. SINTECE generates invoice and processes payment
5. Certificate issued with amendment/cancellation capabilities
6. Status updates communicated via API polling

Amendment Flow:

1. JUL validates amendment eligibility using canAmend status flag
2. JUL submits amendment request via POST /api/CTNs/{id}/amendments with change tracking
3. SINTECE validates amendment request and notifies ARCCLA broker
4. ARCCLA broker reviews amendment in SINTECE with impact analysis
5. SINTECE sends amendment approval/rejection to JUL via notifications
6. If approved, SINTECE updates certificate and generates amendment invoice if applicable
7. JUL processes any additional payments and updates certificate status

Cancellation Flow:

1. JUL validates cancellation eligibility using canCancel status flag
2. JUL submits cancellation request via POST /api/CTNs/{id}/cancellations with reason documentation
3. SINTECE validates cancellation request and calculates financial impact
4. ARCCLA broker reviews cancellation in SINTECE with impact analysis
5. SINTECE sends cancellation approval/rejection to JUL via notifications
6. If approved, SINTECE processes cancellation including any refunds and updates certificate status

2.3 System Actors

The following actors interact with the integrated system:

Actor	Role	Responsibilities	System Access
Trader (Importer/Exporter)	Business entity shipping goods to/from Angola	• Initiates CNCA certificate requests • Uploads required documents (BL, DUP) • Nominates customs broker • Reviews and approves certificate details • Makes payment for certificate issuance • Downloads issued certificates • Views amendment/cancellation history and status	JUL System (Web Portal)
Customs Broker / Freight Forwarder	Licensed agent representing trader with amendment/cancellation capabilities	• Accepts nomination from trader • Completes certificate application • Submits request for approval • Creates amendment requests with change tracking • Creates cancellation requests with reason documentation • Tracks amendment/cancellation status • Receives and forwards certificates to trader • Handles communication with authorities	JUL System (Web Portal with Amendment/Cancellation Features)

ARCCLA Broker	Government official authorized to approve certificates, amendments, and cancellations	• Reviews submitted certificate requests • Validates data accuracy and completeness • Approves or rejects requests with detailed reasoning • Reviews and approves/rejects amendment requests • Reviews and approves/rejects cancellation requests • Performs impact analysis for amendments/cancellations • Provides rejection reasons/comments • Issues official CNCA certificates • Monitors compliance	SINTECE System (Internal Portal with Amendment/Cancellation Workflows)
System Administrator	Technical staff managing system operations	• User management and access control with role-based permissions • System configuration including amendment/cancellation workflows • Monitoring and troubleshooting • Data backup and recovery including amendment/cancellation audit trails • Performance optimization	JUL & SINTECE Systems (Admin Interface)
Payment Processor	Financial institution/payment gateway with enhanced capabilities	• Processes payment transactions including amendment fees • Provides payment confirmations • Enhanced: Handles refunds for approved cancellations • Enhanced: Manages amendment fee processing • Supports multiple currencies and payment methods	Payment Gateway System (Enhanced Integration)

Compliance Officer	Regulatory oversight with enhanced monitoring capabilities	• Enhanced: Monitors amendment/cancellation patterns for compliance • Enhanced: Reviews audit trails for regulatory compliance • Enhanced: Generates compliance reports including amendment/cancellation metrics • Ensures adherence to ARCCLA regulations	SINTECE System (Enhanced Compliance Dashboard)
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Enhanced User Interaction Flow:

Standard Certificate Flow:

1. **Trader** logs into enhanced JUL → Creates certificate request → Uploads documents → Nominates broker
2. **Customs Broker** logs into enhanced JUL → Accepts nomination → Completes application with real-time validation → Submits to SINTECE
3. **ARCCLA Broker** logs into enhanced SINTECE → Reviews request with decision support tools → Approves/Rejects → Generates invoice (if approved)
4. **Customs Broker** receives notification in JUL → Reviews invoice → Processes payment
5. **ARCCLA Broker** receives payment confirmation → Issues certificate with amendment/cancellation capabilities
6. **Trader** and **Customs Broker** receive certificate in JUL → Download and use for customs clearance

Amendment Flow:

1. **Customs Broker** checks amendment eligibility in JUL → Creates amendment request → Documents changes
2. **ARCCLA Broker** receives amendment notification in SINTECE → Reviews changes with impact analysis → Approves/Rejects
3. **Customs Broker** receives amendment decision in JUL → Processes any additional payments → Updates stakeholders
4. **Trader** receives updated certificate with amendment history in JUL

Cancellation Flow:

1. **Customs Broker** checks cancellation eligibility in JUL → Creates cancellation request → Documents reasons
2. **ARCCLA Broker** receives cancellation notification in SINTECE → Reviews with financial impact analysis → Approves/Rejects
3. **Customs Broker** receives cancellation decision in JUL → Processes any refunds → Updates stakeholders
4. **Trader** receives cancellation confirmation and any applicable refunds

2.4 Amendment and Cancellation Capabilities

Amendment Management:

- Real-time eligibility validation based on certificate status
- Change tracking with before/after value comparison
- Impact analysis for proposed amendments
- Role-based approval workflow
- Amendment fee calculation and payment processing
- Complete audit trail with history

Cancellation Management:

- Real-time eligibility validation with financial impact assessment
- Reason code documentation
- Automated refund calculation based on cancellation timing
- Role-based approval workflow with escalation
- Integration with payment systems for refund processing
- Complete audit trail with financial reconciliation

Enhanced Status Management:

- Dynamic eligibility flags (canAmend, canCancel) based on real-time business rules
- Enhanced status workflow supporting amendment/cancellation states
- Real-time status updates with webhook notifications
- Comprehensive status history with audit trail
- Status-based UI enhancements for improved user experience

Advanced Validation Framework:

- Multi-level validation (field-level, business rule, cross-system)
 - Real-time validation with immediate feedback
 - Configurable validation rules with role-based exceptions
 - Comprehensive error messaging with corrective action guidance
 - Integration testing validation for amendment/cancellation workflows
-

3. Process Flows

3.1 As-Is Process Analysis

The current SINTECE workflow consists of 15 distinct steps (LC-AR-CNCA-01 through LC-AR-CNCA-15) that involve multiple manual handoffs between traders, customs brokers, and ARCCLA staff.

3.1.1 Current State Process Flow Diagram

LC-AR-CNCA-01: Trader Issues DUP Certificate



LC-AR-CNCA-02: Trader Receives DUP & BL



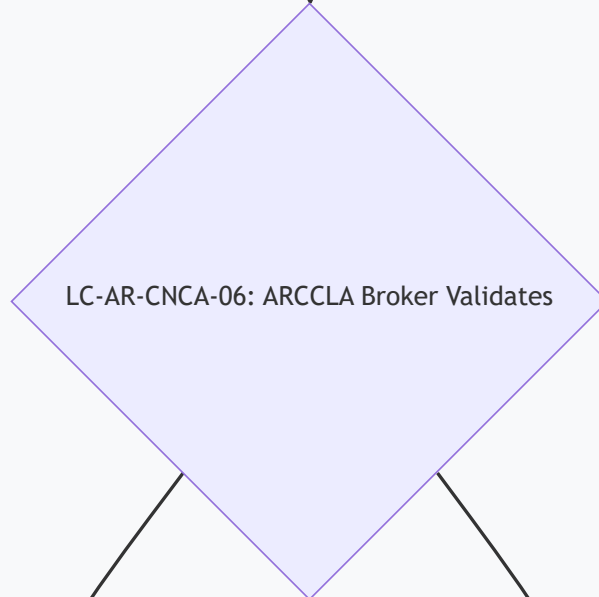
LC-AR-CNCA-03: Customs Broker Receives DUP & BL



LC-AR-CNCA-04: Customs Broker Initiates CNCA Request



LC-AR-CNCA-05: Submit DUP & BL, Assign ARCCLA Broker



Approve



LC-AR-CNCA-07: Generate Invoice & CNCA Draft



LC-AR-CNCA-08: Customs Broker Receives Invoice

Reject



Request Rejected - End

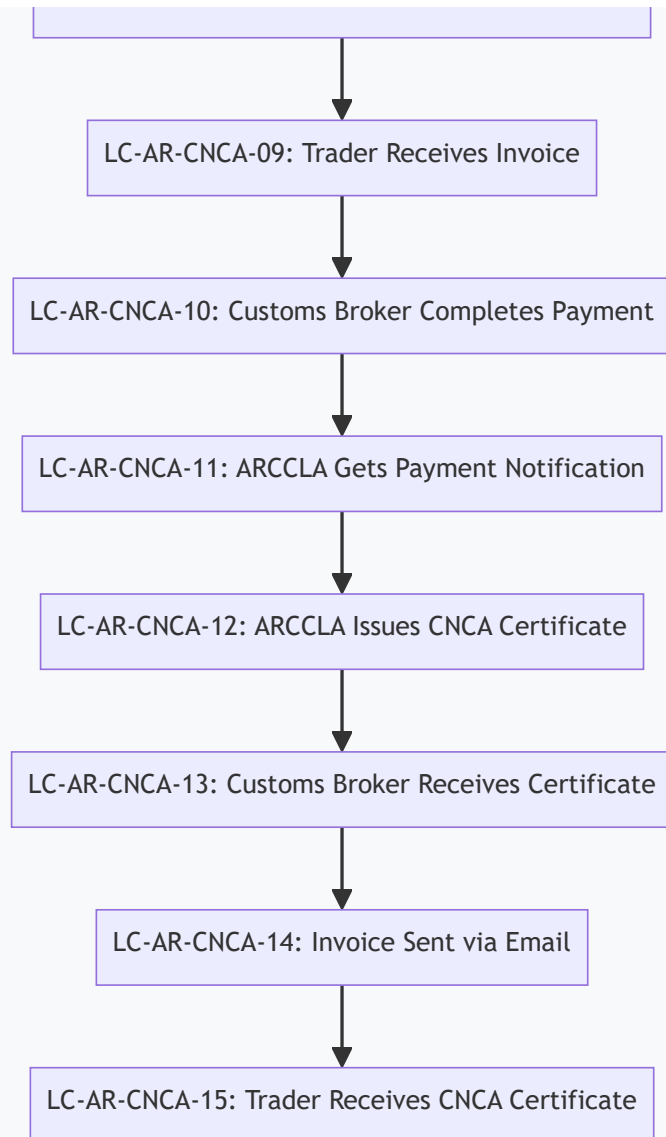


Figure 3.1: Traditional SINTECE Workflow Process

 Process Inefficiencies:

- Multiple email-based communications causing delays
- Manual document handling and verification processes
- Limited real-time visibility for stakeholders
- Duplicate data entry requirements across systems
- No automated amendment or cancellation capabilities

 Key Bottlenecks:

- ARCCLA broker manual review and approval steps
- Payment processing and reconciliation delays

- Physical document submission requirements
- Manual invoice generation and distribution

3.2 To-Be Process

The JUL-SINTECE integration introduces a fully digital workflow with improvements over the current process.

3.2.1 Future State Process Flow Diagram

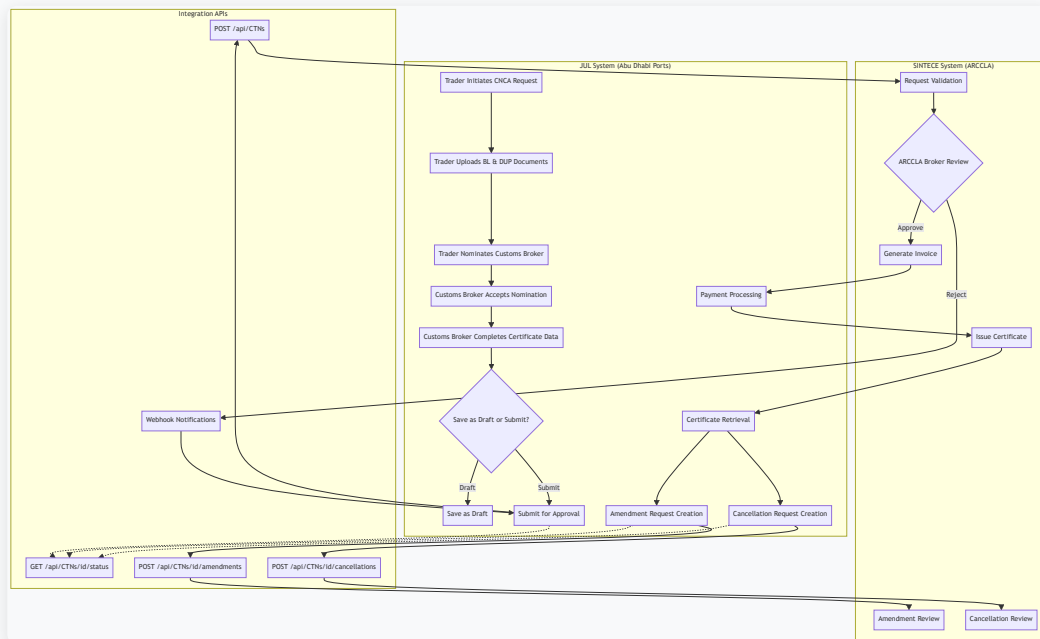


Figure 3.2: Enhanced Future State Core Certificate Workflow

✓ Core Certificate Process Improvements:

1. **Digital Initiation:** Traders initiate requests directly through JUL system
2. **Document Upload:** Electronic submission of BL and DUP documents
3. **Automated Validation:** Real-time validation of submitted data
4. **Streamlined Approval:** ARCCLA brokers receive structured requests for review
5. **Integrated Payment:** Seamless payment processing with automatic reconciliation
6. **Digital Certificate Issuance:** Electronic certificate generation and delivery

🚀 Enhanced Capabilities Integration:

- Real-time amendment requests with ARCCLA approval workflow
- Cancellation management with refund processing capabilities
- Status polling with eligibility flags for allowed operations
- Comprehensive audit trail and notifications system

Benefits of Enhanced Process:

- Fully digital workflow with no physical documents
- Real-time status visibility for all parties
- Integrated amendment and cancellation capabilities
- Automated payment processing
- Single data entry point
- Complete audit trail

3.3 Integration Touchpoints

The integration establishes comprehensive touchpoints between JUL and SINTECE systems with a modern API-driven architecture.

3.3.1 API Integration Architecture Diagram



- Master data synchronization (countries, ports, cargo types)
- User account and role management coordination
- Certificate status and lifecycle management
- Amendment and cancellation request handling

- Authentication and authorization endpoints
- Certificate submission and retrieval APIs
- Amendment request creation and approval APIs
- Cancellation request creation and approval APIs
- Status monitoring and notification APIs

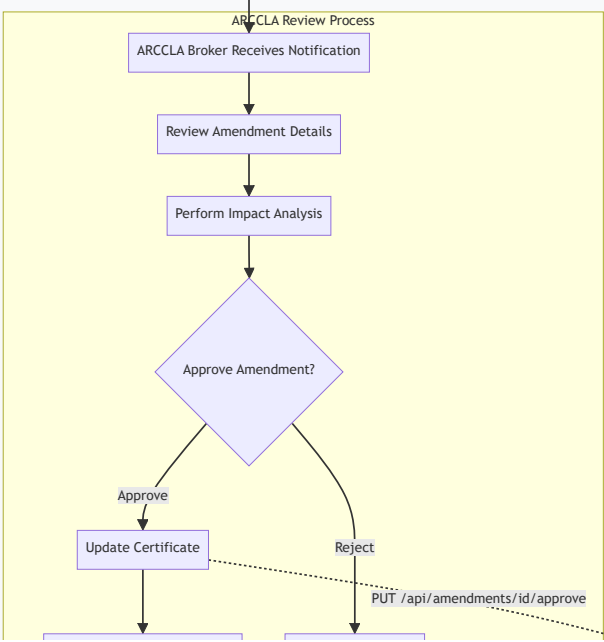
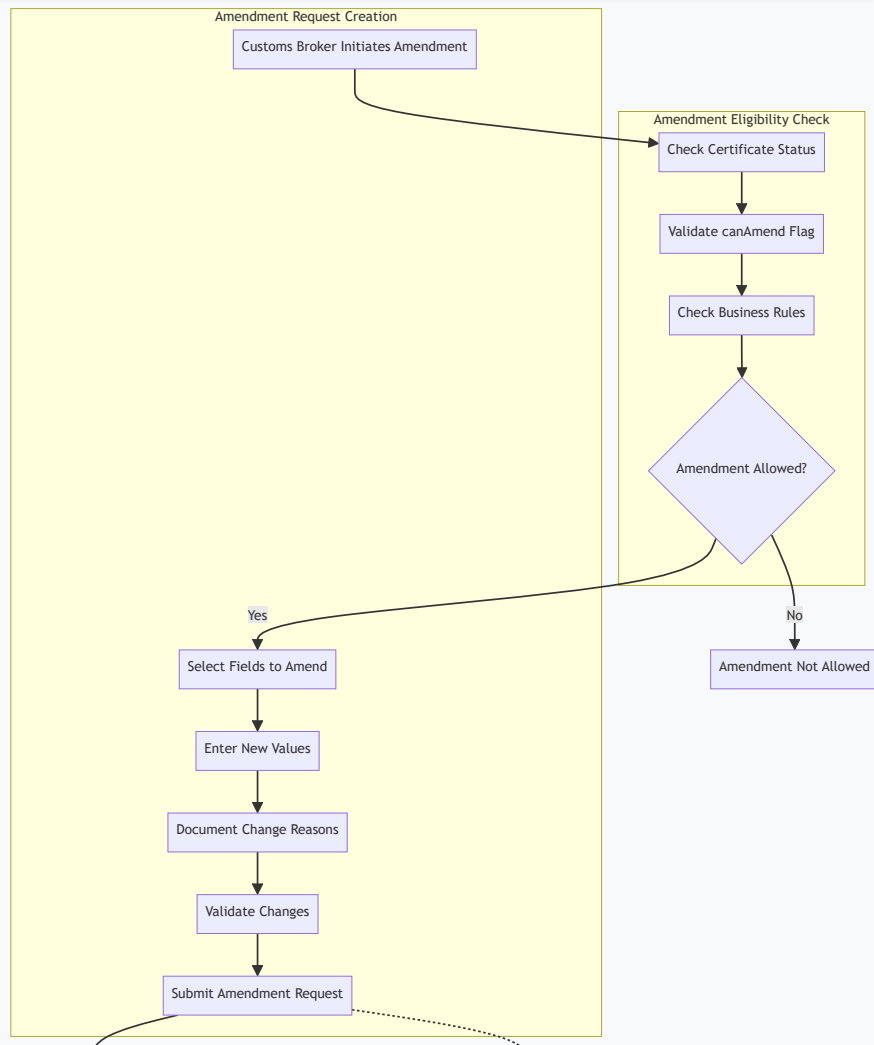
Business Process Touchpoints:

- ARCCLA broker approval workflows
- Payment processing and invoice generation
- Document management and storage
- Audit logging and compliance reporting

3.4 Amendment Workflow Process

The amendment workflow provides controlled modification capabilities for issued certificates with comprehensive ARCCLA approval processes.

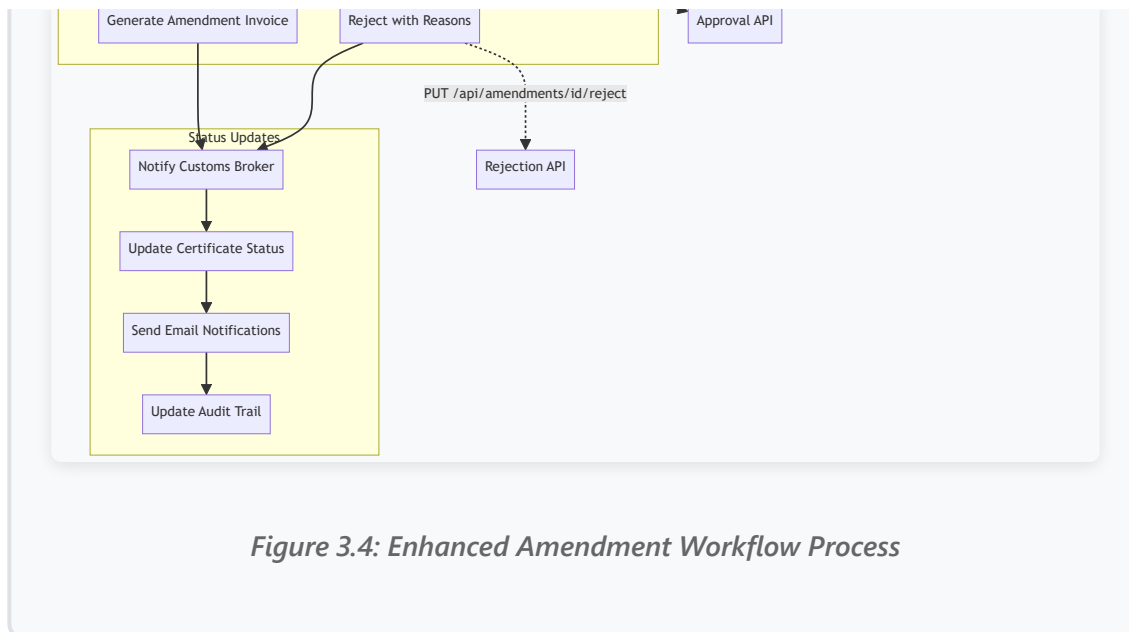
3.4.1 Enhanced Amendment Workflow Diagram



POST /api/CTNs/id/amendments

Amendment API

PUT /api/amendments/id/approve



Amendment Request Initiation:

1. **Eligibility Check:** System validates certificate status and amendment permissions
2. **Request Creation:** Customs broker selects fields to amend and provides change reasons
3. **Validation:** System performs business rule validation on proposed changes
4. **Submission:** Amendment request is submitted to ARCCLA for review

ARCCLA Review and Approval:

1. **Notification:** ARCCLA broker receives amendment request notification
2. **Impact Analysis:** Review of proposed changes and potential impacts
3. **Decision:** Approval or rejection with detailed reasoning
4. **Processing:** For approved amendments, certificate is updated and new invoice generated

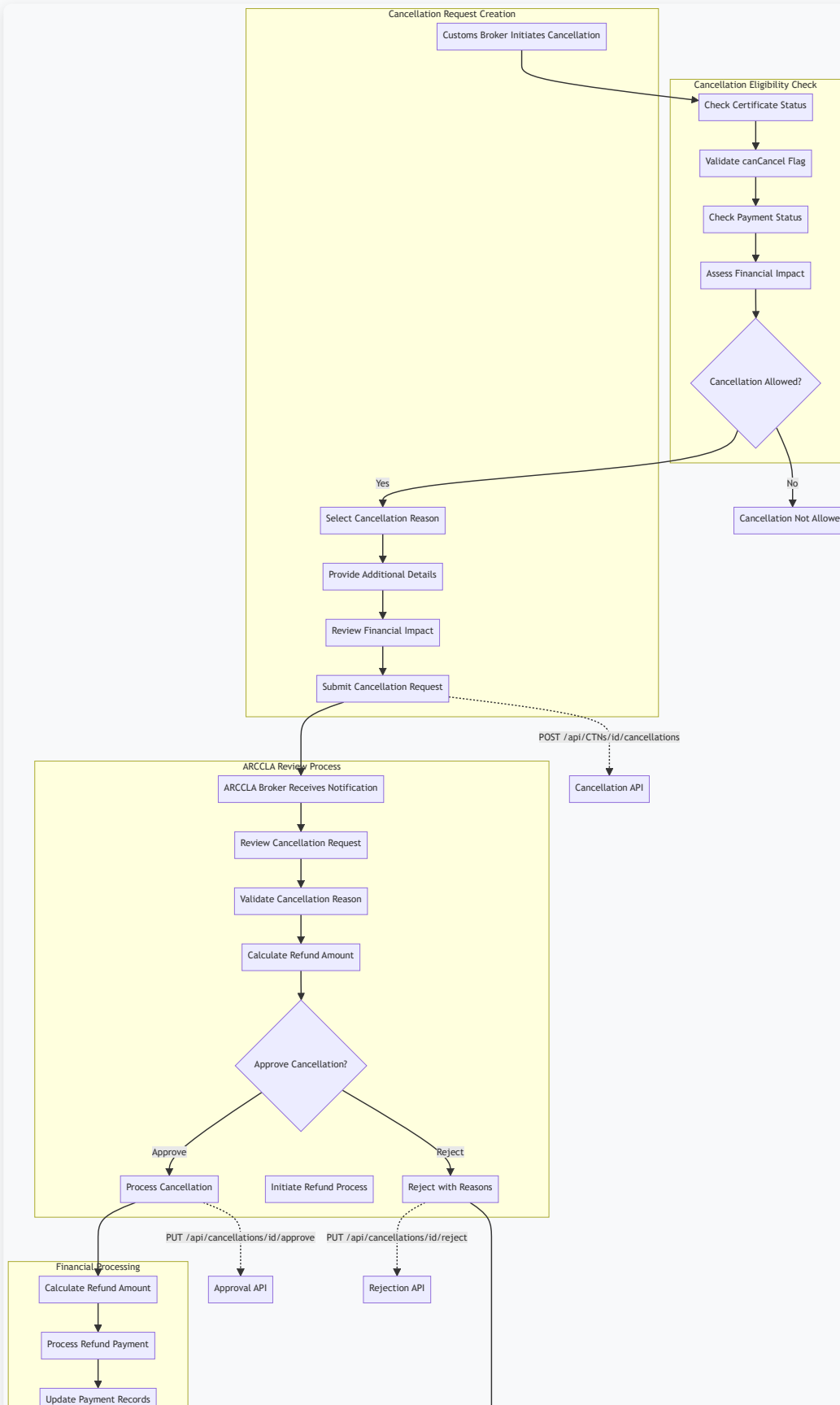
Amendment Eligibility Rules:

- Certificate status must be "Submitted", "Approved", or "Issued"
- No pending amendments or cancellations exist
- Certificate not yet used for customs clearance
- Amendment requested within allowed timeframe (typically 24 hours after issuance)
- User has appropriate amendment permissions

3.5 Cancellation Workflow Process

The cancellation workflow provides controlled termination of certificate requests with comprehensive financial impact management and refund processing.

3.5.1 Enhanced Cancellation Workflow Diagram



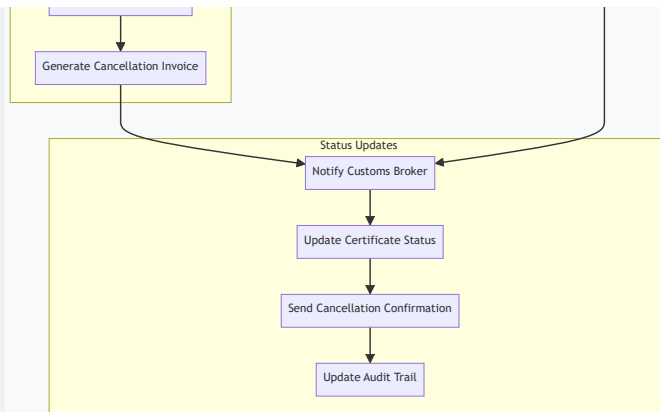


Figure 3.5: Enhanced Cancellation Workflow Process

Cancellation Request Initiation:

1. **Eligibility Check:** System validates certificate status and cancellation permissions
2. **Impact Assessment:** System calculates financial impact and refund eligibility
3. **Request Creation:** Customs broker provides cancellation reason and additional details
4. **Submission:** Cancellation request is submitted to ARCCLA for review

ARCCLA Review and Processing:

1. **Notification:** ARCCLA broker receives cancellation request notification
2. **Review:** Validation of cancellation reason and business justification
3. **Financial Analysis:** Calculation of appropriate refund amount based on timing and status
4. **Decision:** Approval or rejection of cancellation request
5. **Processing:** For approved cancellations, certificate is cancelled and refund initiated

Cancellation Eligibility Rules:

- Certificate status allows cancellation (not yet issued or used for customs clearance)
- Payment status allows refund processing
- Cancellation requested within allowed timeframe
- No active amendments in progress
- Valid business reason for cancellation provided

3.6 ARCCLA Approval Workflows

The ARCCLA approval workflows are enhanced to support the new amendment and cancellation capabilities:

Enhanced Review Dashboard:

- Unified queue for certificate submissions, amendments, and cancellations
- Priority-based processing with configurable business rules
- Rich context information for informed decision making
- Integrated communication tools for broker-to-broker coordination

Approval Decision Framework:

- Structured decision criteria for different request types

- Automated recommendations based on historical patterns
- Escalation procedures for complex or high-value requests
- Audit trail capture for all approval decisions

Workflow Automation:

- Automatic assignment of requests to available brokers
- SLA monitoring with escalation triggers
- Batch processing capabilities for high-volume periods
- Integration with existing SINTECE approval processes

Enhanced Notification System:

- Real-time notifications to all stakeholders
 - Email and system notifications with rich content
 - Webhook support for external system integration
 - Configurable notification preferences by user role
-

4. Data Models

4.1 Core CTN Entity Structure

The Certificate Tracking Note (CTN) entity serves as the foundation for all certificate operations. The enhanced data model includes additional fields to support amendment and cancellation workflows:

```
{
  "id": "string (UUID)",
  "certificateNumber": "string (auto-generated)",
  "requestDate": "datetime (ISO 8601)",
  "submissionDate": "datetime (ISO 8601)",
  "status": "string (enum)",
  "canAmend": "boolean",
  "canCancel": "boolean",
  "lastAmendmentDate": "datetime (ISO 8601, nullable)",
  "lastCancellationAttempt": "datetime (ISO 8601, nullable)",
  "isAmended": "boolean (default: false)",
  "isCancelled": "boolean (default: false)",
  "originalCertificateId": "string (UUID, nullable)",
  "trader": {
    "traderId": "string (UUID)",
    "traderName": "string (required, max 200)",
    "traderEmail": "string (required, valid email)",
    "traderPhone": "string (required)",
    "traderAddress": {
      "street": "string (required, max 200)",
      "city": "string (required, max 100)",
      "country": "string (required, ISO country code)",
      "postalCode": "string (optional, max 20)"
    }
  },
  "customsBroker": {
    "brokerId": "string (UUID)",
    "brokerName": "string (required, max 200)",
    "brokerEmail": "string (required, valid email)",
    "brokerLicenseNumber": "string (required, max 50)",
    "brokerCompany": "string (required, max 200)",
    "isNominated": "boolean (default: false)",
    "nominationDate": "datetime (ISO 8601, nullable)"
  },
  "shipment": {
    "billOfLadingNumber": "string (required, max 50)",
    "dupNumber": "string (required, max 50)",
    "vesselName": "string (required, max 200)",
    "voyageNumber": "string (required, max 50)",
    "portOfLoading": "string (required, port code)",
    "portOfDischarge": "string (required, port code)",
    "estimatedTimeOfArrival": "datetime (ISO 8601)",
    "actualTimeOfArrival": "datetime (ISO 8601, nullable)"
  },
  "goods": [
    {
      "itemNumber": "integer (sequence)",
      "description": "string (required, max 500)",
      "hsCode": "string (required, HS code format)",
      "quantity": "number (required, positive)",
      "unit": "string (required, standard units)",
      "weight": "number (required, positive, kg)",
      "volume": "number (optional, positive, m³)"
    }
  ]
}
```



```

        "value": "number (required, positive, USD)",
        "origin": "string (required, ISO country code)",
        "manufacturer": "string (optional, max 200)"
    }
],
"containers": [
    {
        "containerNumber": "string (required, max 20)",
        "containerType": "string (required, ISO container type)",
        "sealNumbers": ["string (max 50)"],
        "tareWeight": "number (required, positive, kg)",
        "grossWeight": "number (required, positive, kg)",
        "cargoWeight": "number (required, positive, kg)"
    }
],
"payment": {
    "invoiceNumber": "string (auto-generated)",
    "amount": "number (required, positive, USD)",
    "currency": "string (required, ISO currency code)",
    "paymentStatus": "string (enum)",
    "paymentDate": "datetime (ISO 8601, nullable)",
    "paymentMethod": "string (enum)",
    "refundAmount": "number (optional, positive, USD)",
    "refundStatus": "string (enum, nullable)",
    "refundDate": "datetime (ISO 8601, nullable)"
},
"audit": {
    "createdBy": "string (UUID)",
    "createdDate": "datetime (ISO 8601)",
    "lastModifiedBy": "string (UUID)",
    "lastModifiedDate": "datetime (ISO 8601)",
    "version": "integer (auto-increment)"
}
}

```

Key Enhancements:

- Added `canAmend` and `canCancel` eligibility flags
- Enhanced payment model with refund capabilities
- Added amendment and cancellation tracking fields
- Comprehensive audit trail with versioning

4.2 Amendment Entity Structure

The amendment entity captures all modification requests with complete change tracking:

```

{
    "amendmentId": "string (UUID)",
    "originalCertificateId": "string (UUID, foreign key)",
    "amendmentNumber": "string (auto-generated, sequential)",
    "requestDate": "datetime (ISO 8601)",
    "submittedBy": "string (UUID, foreign key)",
    "submittedByRole": "string (enum: CustomsBroker, Trader)",
    "status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
    "requestReason": "string (required, max 500)",
    "businessJustification": "string (optional, max 1000)",
    "requestedChanges": [
        {
            "fieldPath": "string (required, JSON path notation)",
            "fieldName": "string (required, human-readable)",

```

```

        "oldValue": "any (original value)",
        "newValue": "any (proposed new value)",
        "changeReason": "string (required, max 200)",
        "validationStatus": "string (enum: Valid, Invalid, Warning)"
    }
},
"arcclaReview": {
    "assignedTo": "string (UUID, nullable)",
    "assignedDate": "datetime (ISO 8601, nullable)",
    "reviewStarted": "datetime (ISO 8601, nullable)",
    "reviewCompleted": "datetime (ISO 8601, nullable)",
    "decision": "string (enum: Approved, Rejected, nullable)",
    "decisionReason": "string (optional, max 1000)",
    "reviewComments": "string (optional, max 2000)",
    "impactAssessment": {
        "riskLevel": "string (enum: Low, Medium, High)",
        "affectedSystems": ["string"],
        "complianceImpact": "string (optional, max 500)",
        "financialImpact": "number (optional, USD)"
    }
},
"fees": {
    "amendmentFee": "number (required, positive, USD)",
    "currency": "string (required, ISO currency code)",
    "calculationMethod": "string (required)",
    "invoiceNumber": "string (auto-generated, nullable)",
    "paymentStatus": "string (enum)",
    "paymentDate": "datetime (ISO 8601, nullable)"
},
"processing": {
    "applicationStarted": "datetime (ISO 8601, nullable)",
    "applicationCompleted": "datetime (ISO 8601, nullable)",
    "certificateUpdated": "datetime (ISO 8601, nullable)",
    "notificationsSent": "datetime (ISO 8601, nullable)",
    "processingErrors": [
        {
            "errorCode": "string",
            "errorMessage": "string",
            "errorDate": "datetime (ISO 8601)"
        }
    ]
},
"audit": {
    "createdBy": "string (UUID)",
    "createdDate": "datetime (ISO 8601)",
    "lastModifiedBy": "string (UUID)",
    "lastModifiedDate": "datetime (ISO 8601)",
    "workflowHistory": [
        {
            "action": "string",
            "performedBy": "string (UUID)",
            "performedDate": "datetime (ISO 8601)",
            "comments": "string (optional)",
            "systemGenerated": "boolean"
        }
    ]
}
}

```

4.3 Cancellation Entity Structure

The cancellation entity manages certificate termination requests with comprehensive financial tracking:

```
{
  "cancellationId": "string (UUID)",
  "originalCertificateId": "string (UUID, foreign key)",
  "cancellationNumber": "string (auto-generated, sequential)",
  "requestDate": "datetime (ISO 8601)",
  "submittedBy": "string (UUID, foreign key)",
  "submittedByRole": "string (enum: CustomsBroker, Trader)",
  "status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
  "cancellationReason": {
    "reasonCode": "string (enum: required)",
    "reasonDescription": "string (required, max 500)",
    "additionalDetails": "string (optional, max 1000)",
    "urgencyLevel": "string (enum: Normal, High, Critical)"
  },
  "financialImpact": {
    "originalAmount": "number (required, USD)",
    "refundEligibility": "boolean",
    "refundAmount": "number (optional, USD)",
    "cancellationFee": "number (optional, USD)",
    "netRefund": "number (optional, USD)",
    "refundCalculationMethod": "string (required)",
    "refundJustification": "string (optional, max 500)"
  },
  "arcclaReview": {
    "assignedTo": "string (UUID, nullable)",
    "assignedDate": "datetime (ISO 8601, nullable)",
    "reviewStarted": "datetime (ISO 8601, nullable)",
    "reviewCompleted": "datetime (ISO 8601, nullable)",
    "decision": "string (enum: Approved, Rejected, nullable)",
    "decisionReason": "string (optional, max 1000)",
    "reviewComments": "string (optional, max 2000)",
    "riskAssessment": {
      "fraudRisk": "string (enum: Low, Medium, High)",
      "complianceRisk": "string (enum: Low, Medium, High)",
      "operationalRisk": "string (enum: Low, Medium, High)",
      "overallRisk": "string (enum: Low, Medium, High)"
    }
  },
  "refundProcessing": {
    "refundInitiated": "datetime (ISO 8601, nullable)",
    "refundCompleted": "datetime (ISO 8601, nullable)",
    "refundMethod": "string (enum: nullable)",
    "refundReference": "string (nullable)",
    "refundStatus": "string (enum: nullable)",
    "processingErrors": [
      {
        "errorCode": "string",
        "errorMessage": "string",
        "errorDate": "datetime (ISO 8601)"
      }
    ]
  },
  "audit": {
    "createdBy": "string (UUID)",
    "createdDate": "datetime (ISO 8601)",
    "lastModifiedBy": "string (UUID)",
    "lastModifiedDate": "datetime (ISO 8601)",
    "workflowHistory": [
      {

```

```

        "action": "string",
        "performedBy": "string (UUID)",
        "performedDate": "datetime (ISO 8601)",
        "comments": "string (optional)",
        "systemGenerated": "boolean"
    }
]
}
}

```

4.4 Enhanced Status Management

The enhanced status management system provides comprehensive lifecycle tracking with dynamic eligibility determination:

Certificate Status Enumeration:

```

{
  "certificateStatuses": [
    "Draft",
    "Submitted",
    "UnderReview",
    "RequiresAmendment",
    "Approved",
    "PaymentPending",
    "PaymentReceived",
    "Issued",
    "Active",
    "Amended",
    "Cancelled",
    "Expired",
    "Rejected"
  ]
}

```

Amendment Status Enumeration:

```

{
  "amendmentStatuses": [
    "Pending",
    "UnderReview",
    "RequiresInformation",
    "Approved",
    "Rejected",
    "PaymentPending",
    "PaymentReceived",
    "Processing",
    "Completed",
    "Cancelled",
    "Expired"
  ]
}

```

Cancellation Status Enumeration:

```
{
  "cancellationStatuses": [
    "Pending",
    "UnderReview",
    "RequiresInformation",
    "Approved",
    "Rejected",
    "RefundPending",
    "RefundProcessing",
    "RefundCompleted",
    "Completed",
    "Cancelled"
  ]
}
```

Eligibility Rules Matrix: The system dynamically calculates eligibility flags based on the following business rules:

Amendment Eligibility (`canAmend`):

- Certificate status is in: ["Submitted", "Approved", "Issued", "Active"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was issued within the allowed amendment timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Cancellation Eligibility (`canCancel`):

- Certificate status is in: ["Submitted", "Approved", "PaymentPending", "PaymentReceived", "Issued"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was submitted within the allowed cancellation timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Dynamic Status Calculation: The system recalculates eligibility flags whenever:

- Certificate status changes
 - Amendment or cancellation requests are submitted
 - Time-based rules expire
 - User permissions are modified
 - Business rules are updated by administrators
-

5. Core API Specifications

5.1 Master Data APIs

Master data APIs provide essential reference data required for certificate creation, validation, and business operations.

5.1.1 Cargo Types API

Business Purpose: Manages cargo classification data essential for determining certificate fees, validation rules, and compliance requirements. Critical for proper cargo categorization in international shipping, affecting customs documentation, handling procedures, insurance requirements, and regulatory compliance.

Endpoint: `GET /api/CargoTypes`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	CargoType_Desc
2	CargoType_Desc	String - 200	O	Filter by cargo type description	CONTAINER
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **CARGO_UI_001:** CargoType_Desc filter optional, 1-200 characters
- **CARGO_UI_002:** Sort parameter from predefined fields only
- **CARGO_UI_003:** Max 100 records per page

Business Validation Rules:

- **CARGO_BV_001:** Only active cargo types can be used for new CTN creation
- **CARGO_BV_002:** Container cargo requires container details
- **CARGO_BV_003:** Dangerous goods cargo requires special documentation

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique cargo type identifier	1
2	CargoType_Code	String - 10	M	Cargo type code	CONT
3	CargoType_Desc	String - 200	M	Cargo type description	CONTAINER
4	Active	Boolean	M	Active status indicator	true

5	CreatedOn	DateTime	O	Record creation timestamp	2023-01-15T10:30:00Z
---	-----------	----------	---	---------------------------	----------------------

Error Codes:

- **CARGO_E001:** "Invalid parameters" (HTTP 400)
- **CARGO_E002:** "No data found" (HTTP 404)
- **CARGO_E003:** "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/CargoTypes?$sort=CargoType_Desc&CargoType_Desc=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "CargoType_Code": "CONT",
    "CargoType_Desc": "CONTAINER",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "CargoType_Code": "BULK",
    "CargoType_Desc": "BULK CARGO",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.2 Incoterms API

Business Purpose: Provides International Commercial Terms (Incoterms) data crucial for determining responsibility boundaries, costs, and risks between buyers and sellers in international trade transactions. Affects shipping responsibilities, insurance requirements, customs clearance obligations, and liability allocation throughout the shipping process.

Endpoint: GET /api/Incoterms

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	IncotermCode

2	IncotermCode	String - 10	O	Filter by Incoterm code	CIF
3	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **INCO_UI_001:** IncotermCode filter optional, exactly 3 uppercase letters
- **INCO_UI_002:** Display code and description together
- **INCO_UI_003:** Default to "Active Only" for new certificates

Business Validation Rules:

- **INCO_BV_001:** CIF/CIP terms require insurance documentation
- **INCO_BV_002:** Some Incoterms not applicable for certain cargo types
- **INCO_BV_003:** Incoterm selection affects customs valuation

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique Incoterm identifier	2
2	IncotermCode	String - 10	M	Incoterm code	CIF
3	IncotermDesc	String - 200	M	Incoterm description	Cost, Insurance and Freight
4	Active	Boolean	M	Active status indicator	true

Error Codes:

- **INCO_E001:** "Invalid format" (HTTP 400)
- **INCO_E002:** "Not applicable for cargo type" (HTTP 422)
- **INCO_E003:** "Inactive Incoterm" (HTTP 422)

Sample JSON Response:

```
[
  {
    "Id": 2,
    "IncotermCode": "CIF",
    "IncotermDesc": "Cost, Insurance and Freight",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedBy": 1
  }
]
```

5.1.3 Countries API

Business Purpose: Essential for international trade compliance, determining applicable regulations, duties, restrictions, and routing requirements for cargo shipments. Critical for sanctions screening, trade agreement benefits,

customs procedures, and ensuring compliance with import/export regulations between trading nations.

Endpoint: GET /api/Countries

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Country_Name
2	Country_Name	String - 100	O	Filter by country name	Angola
3	ExportingCountry	Boolean	O	Filter exporting countries	true
4	active	Boolean	O	Filter active records only	1

UI Validation Rules:

- **COUNTRY_UI_001:** Country name filter optional, 1-100 characters
- **COUNTRY_UI_002:** Auto-complete dropdown with country name and code
- **COUNTRY_UI_003:** Visual indicator for sanctioned/restricted countries

Business Validation Rules:

- **COUNTRY_BV_001:** Angola must be origin or destination for CNCA certificates
- **COUNTRY_BV_002:** Sanctioned countries require special authorization
- **COUNTRY_BV_003:** Country code must match ISO 3166-1 standard

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique country identifier	10
2	Country_Name	String - 100	M	Country name	Angola
3	Country_Code	String - 3	M	ISO country code	AO
4	ExportingCountry	Boolean	M	Exporting country flag	true
5	ImportingCountry	Boolean	M	Importing country flag	false

Error Codes:

- **COUNTRY_E001:** "Invalid format" (HTTP 400)
- **COUNTRY_E002:** "Not approved for trade" (HTTP 422)
- **COUNTRY_E003:** "Sanctioned country" (HTTP 403)

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Country_Name

2	Country_Name	String - 100	O	Filter by country name	Angola
3	ExportingCountry	Boolean	O	Filter exporting countries	true
4	active	Boolean	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique country identifier	10
2	Country_Name	String - 100	M	Country name	Angola
3	Country_Code	String - 3	M	ISO country code	AO
4	ExportingCountry	Boolean	M	Exporting country flag	true
5	ImportingCountry	Boolean	M	Importing country flag	false

5.1.4 Carriers API

Endpoint: GET /api/Carriers

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 200	O	Filter by carrier name	A.C. ORSSLEFF'S
3	active	Boolean	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique carrier identifier	789
2	Name	String - 200	M	Carrier company name	A.C. ORSSLEFF'S EFTF A/S
3	Code	String - 20	O	Carrier code	ORSSLEFF
4	Active	Boolean	M	Active status indicator	true

5.1.5 Currencies API

Endpoint: GET /api/Currencies

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Code
2	Code	String - 3	O	Filter by currency code	USD
3	active	Boolean	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique currency identifier	2
2	Code	String - 3	M	ISO currency code	USD
3	Name	String - 100	O	Currency name	US Dollar
4	Symbol	String - 5	O	Currency symbol	\$

5.2 Certificate Management APIs

Certificate management APIs handle the core CTN certificate lifecycle operations, enabling the creation, retrieval, and management of CNCA certificates throughout their entire lifecycle from draft to completion.

5.2.1 CTN List API

Business Purpose: Provides comprehensive listing and filtering capabilities for CTN certificates. Essential for dashboard displays, search functionality, status tracking, and bulk operations. Supports real-time monitoring of certificate processing status and enables efficient certificate portfolio management for traders and customs brokers.

Endpoint: GET /api/ctns

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand for related entities	CargoType,Status,Visum_Agent,CreatedBy,Cor
2	\$sort	String - 100	O	OData sort parameter	-ModifiedOn

3	\$top	Integer	O	Number of records to return	10
4	\$filter	String - 1000	O	OData filter expression	Status eq 'Active'

UI Validation Rules:

- **CTN_UI_001:** \$top parameter max 100 records per request
- **CTN_UI_002:** \$expand parameter validate against allowed entity names
- **CTN_UI_003:** \$filter parameter OData syntax validation

Business Validation Rules:

- **CTN_BV_001:** User can only view certificates belonging to their organization
- **CTN_BV_002:** Sensitive data requires role-based access
- **CTN_BV_003:** Date range filters max 2 years for performance

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique CTN identifier	503808
2	CTN_Reference_Number	String - 50	M	System generated CTN reference	AO-CNT-503808-2023
3	BL_number	String - 50	M	Bill of lading number	vc568009iujh
4	UniqueTradeNumber	String - 50	M	Unique trade number (DUP)	56789098765
5	DCNumber	String - 50	O	Declaration certificate number	7777777
6	StatusId	Integer	M	Certificate status identifier	1
7	CargoTypeId	Integer	M	Cargo type identifier	1
8	Total_Value_Of_Goods	Decimal	M	Total value of goods in USD	15000.00
9	CreatedOn	DateTime	M	Record creation timestamp	2025-11-12T18:49:30.88Z
10	ModifiedOn	DateTime	O	Record modification timestamp	2025-11-12T19:15:45.23Z

Error Codes:

- **CTN_E001:** "Invalid parameters" (HTTP 400)
- **CTN_E002:** "Access denied" (HTTP 403)
- **CTN_E003:** "No data found" (HTTP 404)

Sample JSON Response:

```
[
  {
    "Id": 503808,
    "CTN_Reference_Number": "AO-CNT-503808-2023",
    "StatusId": 1,
    "BL_number": "vc568009iujh",
    "UniqueTradeNumber": "56789098765",
    "DCNumber": "7777777",
    "CargoTypeId": 1,
    "ETD": "2023-12-15T10:00:00Z",
    "ETA": "2023-12-20T15:30:00Z",
    "Total_Value_Of_Goods": 15000.00,
    "View_CurrencyId": 2,
    "IsExport": true,
    "IsImport": false,
    "CreatedOn": "2025-11-12T18:49:30.8817831Z",
    "CreatedById": 1,
    "ModifiedOn": "2025-11-12T19:15:45.2345678Z",
    "ModifiedById": 1,
    "Status": {
      "Id": 1,
      "Status_Desc": "Draft",
      "Active": true
    },
    "CargoType": {
      "Id": 1,
      "CargoType_Code": "CONT",
      "CargoType_Desc": "CONTAINER"
    }
  }
]
```

5.2.2 CTN Details API

Endpoint: GET /api/ctns/{id}

Request Elements:

S. No	Attributes	Data Type- Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN unique identifier from URL path	503808
2	\$expand	String - 500	O	OData expand for related entities	CargoType,ParentCTN,Status,Incoterm

Response Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Unique CTN identifier	503808
2	CTN_Reference_Number	String - 50	M	System generated CTN reference	AO-CNT-503808-2023
3	BL_number	String - 50	M	Bill of lading number	vc568009iujh
4	UniqueTradeNumber	String - 50	M	Unique trade number	56789098765
5	VoyageNo	String - 50	O	Voyage number	VOY123
6	VesselName	String - 200	O	Vessel name	MSC MEDITERRANEAN
7	Total_number_containers	Integer	M	Total number of containers	2
8	Total_Value_Of_Goods	Decimal	M	Total value of goods	15000.00
9	CTNCost	Decimal	O	Certificate cost	150.00

5.2.3 CTN Creation API

Endpoint: POST /api/ctns

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	BL_number	String - 50	M	Bill of lading number	vc568009iujh
2	UniqueTradeNumber	String - 50	M	Unique trade number	56789098765
3	CargoTypeId	Integer	M	Cargo type identifier	1
4	IncotermsId	Integer	M	Incoterm identifier	2
5	OriginCountryId	Integer	M	Origin country identifier	10
6	FinalDestinationCountryId	Integer	M	Destination country identifier	85
7	Total_Value_Of_Goods	Decimal	M	Total value of goods	15000.00
8	View_CurrencyId	Integer	M	Currency identifier	2
9	IsExport	Boolean	M	Export flag	true

10	Consigneeld	Integer	O	Consignee identifier	12345
----	-------------	---------	---	----------------------	-------

Sample JSON Request:

```
{
  "BL_number": "vc568009iujh",
  "UniqueTradeNumber": "56789098765",
  "CargoTypeId": 1,
  "IncotermId": 2,
  "OriginCountryId": 10,
  "FinalDestinationCountryId": 85,
  "Total_Value_Of_Goods": 15000.00,
  "View_CurrencyId": 2,
  "IsExport": true,
  "IsImport": false,
  "ETD": "2023-12-15T10:00:00Z",
  "ETA": "2023-12-20T15:30:00Z",
  "VoyageNo": "VOY123",
  "VesselName": "MSC MEDITERRANEAN",
  "ConsigneeId": 12345
}
```

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Generated CTN unique identifier	503809
2	CTN_Reference_Number	String - 50	M	Auto-generated CTN reference	AO-CNT-503809-2023
3	StatusId	Integer	M	Initial status (typically 1 for Draft)	1
4	CreatedOn	DateTime	M	Record creation timestamp	2025-11-13T10:30:00Z
5	CreatedById	Integer	M	User who created the record	13345

5.2.4 Additional Master Data APIs

CTN Cities API

Endpoint: GET /api/CTNCities

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example

1	\$sort	String - 100	O	OData sort parameter	Name
2	CountryId	Integer	M	Country identifier filter	10
3	Name	String - 100	O	City name filter	Luanda
4	active	Boolean	O	Filter active records only	1

Freight Payment Types API

Endpoint: GET /api/FreightPaymentTypes

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	FreightPaymentType_Desc
2	FreightPaymentType_Desc	String - 200	O	Filter by payment type	Prepaid
3	active	Boolean	O	Filter active records only	1

Banks API

Endpoint: GET /api/Banks

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Name
2	Name	String - 200	O	Filter by bank name	Banco Nacional
3	active	Boolean	O	Filter active records only	1

Consignees API

Endpoint: GET /api/Consignees

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	NIFNumber

2	NIFNumber	String - 50	O	Filter by NIF number	123456789
3	active	Boolean	O	Filter active records only	1

Transport Types API

Endpoint: GET /api/TransportTypes

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 100	O	OData expand parameter	description
2	\$sort	String - 100	O	OData sort parameter	Description
3	Description	String - 200	O	Filter by description	Sea
4	active	Boolean	O	Filter active records only	1

Locations/Ports API

Endpoint: GET /api/Locations

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	PortName
2	CountryId	Integer	M	Country identifier filter	4
3	PortName	String - 200	O	Filter by port name	Dubai
4	active	Boolean	O	Filter active records only	1

Goods Classifications API

Endpoint: GET /api/GoodsClassifications

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Goods_Classification_Desc
2	Goods_Classification_Desc	String -	O	Filter by	Electronics

		200		classification	
3	active	Boolean	O	Filter active records only	1

IMO Codes API

Endpoint: GET /api/IMOs

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	IMO_Desc
2	IMO_Desc	String - 200	O	Filter by IMO description	Flammable Liquids
3	active	Boolean	O	Filter active records only	1

Container Types API

Endpoint: GET /api/ContainerTypes

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	Container_Type
2	Container_Type	String - 50	O	Filter by container type	20GP
3	active	Boolean	O	Filter active records only	1

Attachment Names API

Endpoint: GET /api/AttachmentNames

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$sort	String - 100	O	OData sort parameter	AttachmentName_Desc
2	AttachmentName_Desc	String - 200	O	Filter by attachment name	Bill of Lading
3	active	Boolean	O	Filter active records only	1

5.3 CTN Related Entity APIs

5.3.1 CTN Tracking API

Endpoint: GET /api/ctnTracking

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand parameter	CTN, DestinationPort, SourcePort, TransportType
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Tracking record ID	71898
2	CTNId	Integer	M	Related CTN ID	503808
3	SourcePortId	Integer	O	Source port identifier	150
4	DestinationPortId	Integer	O	Destination port identifier	275
5	TransportTypeId	Integer	M	Transport type identifier	1
6	ETD	DateTime	O	Estimated time of departure	2023-12-15T10:00:00Z
7	ETA	DateTime	O	Estimated time of arrival	2023-12-20T15:30:00Z
8	VesselName	String - 200	O	Vessel name	MSC MEDITERRANEAN
9	VoyageNumber	String - 50	O	Voyage number	VOY123456

5.3.2 CTN Attachments API

Endpoint: GET /api/ctnAttachments

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 300	O	OData expand parameter	CreatedBy,attachment,attachmentName
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Attachment record ID	320988
2	CTNId	Integer	M	Related CTN ID	503808
3	AttachmentNameId	Integer	M	Attachment type ID	1
4	AttachmentId	String (UUID)	M	File attachment ID	0da7461c-2155-4d20-a695-6b7463367327
5	FileName	String - 256	M	Original file name	Bill_of_Lading.pdf
6	FileSize	Integer	M	File size in bytes	245760
7	MimeType	String - 100	M	File MIME type	application/pdf
8	CreatedOn	DateTime	M	Upload timestamp	2023-11-12T14:30:00Z

POST /api/ctnAttachments - Create Attachment**Request Elements:**

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	AttachmentNameId	Integer	M	Attachment type identifier	1

3	AttachmentId	String (UUID)	M	File upload identifier	0da7461c-2155-4d20-a695-6b7463367327
4	Description	String - 500	O	Attachment description	Original Bill of Lading

5.3.3 CTN Goods API

Endpoint: GET /api/ctnGoods

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 300	O	OData expand parameter	goodsclassification,cargotype,imo
2	\$sort	String - 100	O	OData sort parameter	gumarref
3	\$with	String - 500	O	OData aggregate functions	sum(grossweightkg),sum(volumecbm)
4	ctn	Integer	M	CTN identifier	503808
5	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Goods record ID	989192
2	CTNId	Integer	M	Related CTN ID	503808
3	Description	String - 500	M	Goods description	Electronic Components
4	HSCode	String - 20	M	Harmonized system code	854390
5	GrossWeightKG	Decimal	M	Gross weight in kilograms	1500.50
6	VolumeCBM	Decimal	O	Volume in cubic meters	12.5

7	ValueOfGoods	Decimal	M	Goods value	15000.00
8	NumberOfPackages	Integer	M	Number of packages	50
9	PackagingType	String - 50	M	Type of packaging	Cartons
10	CountryOfOrigin	String - 50	M	Country of origin	China

POST /api/ctnGoods - Create Goods Record

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	Description	String - 500	M	Goods description	Electronic Components
3	HSCode	String - 20	M	Harmonized system code	854390
4	GrossWeightKG	Decimal	M	Gross weight in kg	1500.50
5	VolumeCBM	Decimal	O	Volume in cubic meters	12.5
6	ValueOfGoods	Decimal	M	Goods value in USD	15000.00
7	NumberOfPackages	Integer	M	Number of packages	50
8	PackagingType	String - 50	M	Packaging type	Cartons
9	CountryOfOriginId	Integer	M	Country of origin ID	45
10	GoodsClassificationId	Integer	O	Goods classification ID	12

5.3.4 CTN Containers API

Endpoint: GET /api/ctnContainers

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	ContainerType
2	ctn	Integer	M	CTN identifier	503808

3	ctnid	Integer	M	CTN identifier (duplicate)	503808
---	-------	---------	---	----------------------------	--------

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Container record ID	510046
2	CTNId	Integer	M	Related CTN ID	503808
3	ContainerNumber	String - 20	M	Container number	MSCU1234567
4	ContainerTypeId	Integer	M	Container type ID	1
5	SealNumbers	String - 200	O	Container seal numbers	SEAL123,SEAL456
6	TareWeightKG	Decimal	M	Tare weight in kg	2300.00
7	GrossWeightKG	Decimal	M	Gross weight in kg	18500.00
8	NetWeightKG	Decimal	M	Net weight in kg	16200.00

POST /api/ctnContainers - Create Container Record

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	ContainerNumber	String - 20	M	Container number	MSCU1234567
3	ContainerTypeId	Integer	M	Container type identifier	1
4	SealNumbers	String - 200	O	Container seal numbers	SEAL123,SEAL456
5	TareWeightKG	Decimal	M	Tare weight in kg	2300.00
6	GrossWeightKG	Decimal	M	Gross weight in kg	18500.00

GET /api/ctnContainers/new - Container Template

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example

1	Id	Integer	M	Template ID (null for new)	null
2	CTNId	Integer	O	CTN ID (to be set)	null
3	ContainerNumber	String - 20	O	Empty container number	""
4	ContainerTypeId	Integer	O	Default container type	1
5	TareWeightKG	Decimal	O	Default tare weight	2300.00

5.3.5 CTN Addresses API

Endpoint: GET /api/ctnAddresses

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	QryAddressType,Country1
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Address record ID	426334
2	CTNId	Integer	M	Related CTN ID	503808
3	AddressTypeId	Integer	M	Address type ID	1
4	CompanyName	String - 200	M	Company name	ABC Trading LLC
5	ContactPerson	String - 100	O	Contact person name	John Smith
6	Address1	String - 200	M	Address line 1	123 Business Street
7	Address2	String - 200	O	Address line 2	Suite 456
8	City	String - 100	M	City	Dubai
9	CountryId	Integer	M	Country identifier	4
10	PostalCode	String - 20	O	Postal/ZIP code	12345

11	Phone	String - 50	O	Phone number	+971-4-1234567
12	Email	String - 200	O	Email address	contact@abc-trading.com

5.3.6 CTN Charges API

Endpoint: GET /api/ctnCharges

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 200	O	OData expand parameter	Charge,ctn.view_currency
2	ctn	Integer	M	CTN identifier	503808
3	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Charge record ID	12345
2	CTNId	Integer	M	Related CTN ID	503808
3	ChargeId	Integer	M	Charge type ID	1
4	Amount	Decimal	M	Charge amount	150.00
5	CurrencyId	Integer	M	Currency identifier	2
6	Description	String - 200	O	Charge description	Certificate Processing Fee
7	IsSystem	Boolean	M	System-generated charge flag	true

5.3.7 User Communication API

Endpoint: GET /api/userCommunication

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example

1	\$expand	String - 100	O	OData expand parameter	modifiedby
2	\$sort	String - 100	O	OData sort parameter	-modifiedon
3	ctn	Integer	M	CTN identifier	503808
4	ctnid	Integer	M	CTN identifier (duplicate)	503808

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Communication record ID	98765
2	CTNId	Integer	M	Related CTN ID	503808
3	Subject	String - 200	M	Communication subject	Document Review Required
4	Message	String - 2000	M	Communication message	Please review attached documents
5	CreatedOn	DateTime	M	Creation timestamp	2023-11-12T16:45:00Z
6	CreatedById	Integer	M	User who created the message	13345
7	IsInternal	Boolean	M	Internal communication flag	false

POST /api/userCommunication - Create Communication

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	CTNId	Integer	M	Related CTN identifier	503808
2	Subject	String - 200	M	Communication subject	Document Review Required
3	Message	String - 2000	M	Communication message	Please review attached documents
4	IsInternal	Boolean	M	Internal communication flag	false
5	RecipientIds	Array	O	Array of recipient user	[13345, 13346]

				IDs	
--	--	--	--	-----	--

5.4 File Management APIs

5.4.1 File Upload API

Endpoint: POST /api/fileupload

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	file	Binary	M	File upload (multipart/form-data)	[binary file data]
2	fileName	String - 256	M	Original file name	Bill_of_Lading.pdf
3	description	String - 500	O	File description	Original shipping document

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fileId	String (UUID)	M	Unique file identifier	0da7461c-2155-4d20-a695-6b7463367327
2	fileName	String - 256	M	Uploaded file name	Bill_of_Lading.pdf
3	fileSize	Integer	M	File size in bytes	245760
4	mimeType	String - 100	M	File MIME type	application/pdf
5	uploadDate	DateTime	M	Upload timestamp	2023-11-12T14:30:00Z

GET /api/fileupload/{fileId} - Download File

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fileId	String (UUID)	M	File identifier from URL path	0da7461c-2155-4d20-a695-6b7463367327

Response Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	contentStream	Base64 Binary	M	Base64 encoded file content	JVBERi0xLjQKJeLjz9MK...
2	fileName	String - 256	M	Original file name	Bill_of_Lading.pdf
3	mimeType	String - 100	M	File MIME type	application/pdf
4	fileSize	Integer	M	File size in bytes	245760

5.6 Specialized CTN APIs

5.6.1 Parent CTN Relationships API

Endpoint: GET /api/Ctns/GetAllowedParentCtns

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	ctnId	Integer	M	Current CTN identifier	503808
2	\$sort	String - 100	O	OData sort parameter	BL_number
3	BL_number	String - 50	O	Filter by Bill of Lading number	MSC123456

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Id	Integer	M	Parent CTN ID	503800
2	CTN_Reference_Number	String - 50	M	Parent CTN reference	AO-CNT-503800-2023
3	BL_number	String - 50	M	Parent Bill of Lading	MSC123456
4	StatusId	Integer	M	Parent CTN status	2
5	CanBeParent	Boolean	M	Eligibility as parent	true

5.6.2 CTN Export API

Endpoint: GET /api/ctns/export

Request Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	\$expand	String - 500	O	OData expand parameter	CargoType,Status,Visum_Agent
2	\$export_format	String - 10	M	Export format	csv
3	\$sort	String - 100	O	OData sort parameter	-ModifiedOn
4	\$top	Integer	O	Number of records	10
5	\$filter	String - 1000	O	OData filter expression	StatusId eq 2

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	Content	Text/CSV	M	CSV formatted CTN data	"Id","CTN_Reference_Number"...
2	ContentType	String - 50	M	Response content type	text/csv
3	FileName	String - 100	M	Suggested file name	CTN_Export_20231112.csv

6. API Capabilities ⚡

6.1 Amendment APIs

Amendment APIs provide controlled modification capabilities for existing certificates with ARCCLA approval workflow.

6.1.1 Amendment Request Creation API

Endpoint: `POST /api/CTNs/{id}/amendments`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	requestReason	String - 500	M	Reason for amendment request	Incorrect vessel name provided
3	businessJustification	String - 1000	O	Detailed business justification	Customer informed of vessel change after booking
4	requestedChanges	Array	M	Array of field changes requested	[field change objects]
5	urgencyLevel	String - 20	O	Urgency level (Normal, High, Critical)	Normal

Requested Changes Array Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	fieldPath	String - 100	M	JSON path to field being changed	VesselName
2	fieldName	String - 100	M	Human-readable field name	Vessel Name
3	oldValue	String - 500	M	Current field value	MSC MEDITERRANEAN
4	newValue	String - 500	M	Proposed new field value	MAERSK ESSEX
5	changeReason	String - 200	M	Specific reason for this field change	Vessel change notification from carrier

Sample JSON Request:

```
{
  "requestReason": "Incorrect vessel name provided",
  "businessJustification": "Customer informed of vessel change after booking confirmation",
  "urgencyLevel": "Normal",
  "requestedChanges": [
    {
      "fieldPath": "VesselName",
      "fieldName": "Vessel Name",
      "oldValue": "MSC MEDITERRANEAN",
      "newValue": "MAERSK ESSEX",
      "changeReason": "Vessel change notification from carrier"
    },
    {
      "fieldPath": "ETA",
      "fieldName": "Estimated Time of Arrival",
      "oldValue": "2023-12-20T15:30:00Z",
      "newValue": "2023-12-22T10:00:00Z",
      "changeReason": "Updated ETA due to vessel change"
    }
  ]
}
```

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Unique amendment identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String - 50	M	Sequential amendment number	AMD-503808-001
3	status	String - 20	M	Amendment status	Pending
4	requestDate	DateTime	M	Amendment request timestamp	2025-11-13T10:45:00Z
5	estimatedFee	Decimal	O	Estimated amendment fee	25.00

Sample JSON Response:

```
{
  "amendmentId": "a1b2c3d4-e5f6-7890-abcd-1234567890ef",
  "amendmentNumber": "AMD-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T10:45:00Z",
  "estimatedFee": 25.00,
  "currency": "USD",
  "submittedBy": 13345,
```

```
"submittedByRole": "CustomsBroker"
}
```

6.1.2 Amendment Status API

Endpoint: GET /api/amendments/{amendmentId}/status

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier from URL path	a1b2c3d4-e5f6-7890-abcd-1234567890ef

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String - 50	M	Amendment number	AMD-503808-001
3	status	String - 20	M	Current amendment status	UnderReview
4	assignedTo	String - 100	O	ARCCLA broker assigned	John.Smith@arccla.ao
5	lastUpdated	DateTime	M	Last status update timestamp	2025-11-13T14:30:00Z
6	statusHistory	Array	O	Amendment status history	[status history objects]

6.1.3 Amendment Approval API

Endpoint: PUT /api/amendments/{amendmentId}/approve

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	amendmentId	String (UUID)	M	Amendment identifier from URL path	a1b2c3d4-e5f6-7890-abcd-1234567890ef

2	decision	String - 20	M	Approval decision (Approved/Rejected)	Approved
3	decisionReason	String - 1000	O	Reason for approval decision	Valid vessel change confirmed
4	reviewComments	String - 2000	O	Additional reviewer comments	Amendment approved without issues
5	amendmentFee	Decimal	O	Final amendment fee	25.00

6.2 Cancellation APIs

Cancellation APIs provide controlled termination of certificate requests with financial impact handling.

6.2.1 Cancellation Request Creation API

Endpoint: `POST /api/CTNs/{id}/cancellations`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	reasonCode	String - 20	M	Standardized cancellation reason code	SHIPMENT_CANCELLED
3	reasonDescription	String - 500	M	Detailed cancellation reason	Shipment cancelled by customer
4	additionalDetails	String - 1000	O	Additional cancellation details	Customer business closure
5	urgencyLevel	String - 20	O	Cancellation urgency (Normal, High, Critical)	High

Sample JSON Request:

```
{
  "reasonCode": "SHIPMENT_CANCELLED",
  "reasonDescription": "Shipment cancelled by customer due to business closure",
  "additionalDetails": "Customer has permanently ceased operations and requires immediate cancellation",
  "urgencyLevel": "High"
}
```

Response Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Unique cancellation identifier	x1y2z3a4-b5c6-7890-defg-9876543210ab
2	cancellationNumber	String - 50	M	Sequential cancellation number	CAN-503808-001
3	status	String - 20	M	Cancellation status	Pending
4	requestDate	DateTime	M	Cancellation request timestamp	2025-11-13T11:30:00Z
5	estimatedRefund	Decimal	O	Estimated refund amount	125.00

Sample JSON Response:

```
{
  "cancellationId": "x1y2z3a4-b5c6-7890-defg-9876543210ab",
  "cancellationNumber": "CAN-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T11:30:00Z",
  "estimatedRefund": 125.00,
  "refundCurrency": "USD",
  "submittedBy": 13345,
  "submittedByRole": "CustomsBroker"
}
```

6.2.2 Cancellation Status API

Endpoint: GET /api/cancellations/{cancellationId}/status

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Cancellation identifier from URL path	x1y2z3a4-b5c6-7890-defg-9876543210ab

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	cancellationId	String (UUID)	M	Cancellation identifier	x1y2z3a4-b5c6-7890-defg-9876543210ab

2	cancellationNumber	String - 50	M	Cancellation number	CAN-503808-001
3	status	String - 20	M	Current cancellation status	UnderReview
4	assignedTo	String - 100	O	ARCCLA broker assigned	Maria.Santos@arccla.ao
5	refundStatus	String - 20	O	Refund processing status	Pending
6	lastUpdated	DateTime	M	Last status update timestamp	2025-11-13T15:45:00Z

6.3 Approval Workflow APIs

Approval workflow APIs enable ARCCLA brokers to review and process amendment/cancellation requests.

6.3.1 Pending Requests API

Endpoint: GET /api/approvals/pending

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	requestType	String - 20	O	Filter by request type (amendments/cancellations)	amendments
2	priority	String - 20	O	Filter by priority level	High
3	assignedTo	String - 100	O	Filter by assigned broker	current_user
4	\$top	Integer	O	Number of records to return	20
5	\$skip	Integer	O	Number of records to skip	0

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	requestId	String (UUID)	M	Amendment/Cancellation identifier	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	requestType	String - 20	M	Type of request (Amendment/Cancellation)	Amendment
3	requestNumber	String - 50	M	Request number	AMD-503808-001

4	certificateId	String - 50	M	Related certificate ID	503808
5	submittedDate	DateTime	M	Request submission date	2025-11-13T10:45:00Z
6	priority	String - 20	M	Request priority	Normal
7	submittedBy	String - 100	M	User who submitted request	CustomsBroker123

6.4 Status Polling APIs

Status polling APIs provide real-time status updates with eligibility flags for allowed operations.

6.4.1 Enhanced Certificate Status API

Endpoint: `GET /api/CTNs/{id}/enhanced-status`

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	id	Integer	M	CTN identifier from URL path	503808
2	includeHistory	Boolean	O	Include status change history	true
3	includeFlags	Boolean	O	Include eligibility flags	true

Response Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	certificateId	Integer	M	Certificate identifier	503808
2	currentStatus	String - 20	M	Current certificate status	Approved
3	statusId	Integer	M	Status identifier	3
4	canAmend	Boolean	M	Amendment eligibility flag	true
5	canCancel	Boolean	M	Cancellation eligibility flag	true
6	lastAmendmentDate	DateTime	O	Date of last amendment	2025-11-10T14:20:00Z
7	pendingAmendments	Integer	M	Number of pending amendments	0

8	pendingCancellations	Integer	M	Number of pending cancellations	0
9	eligibilityReasons	Object	O	Detailed eligibility explanations	{...}

Sample JSON Response:

```
{
  "certificateId": 503808,
  "currentStatus": "Approved",
  "statusId": 3,
  "canAmend": true,
  "canCancel": true,
  "lastStatusUpdate": "2025-11-13T09:30:00Z",
  "lastAmendmentDate": null,
  "pendingAmendments": 0,
  "pendingCancellations": 0,
  "eligibilityReasons": {
    "amendmentEligibility": "Certificate is in approved status and within amendment timeframe",
    "cancellationEligibility": "Certificate not yet issued, cancellation allowed with full refund"
  },
  "businessRules": {
    "amendmentTimeLimit": "24 hours after approval",
    "cancellationTimeLimit": "48 hours after approval",
    "maxAmendmentsAllowed": 3
  }
}
```

6.4.2 Webhook Notifications API

Endpoint: POST /api/webhooks/register

Request Elements:

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	callbackUrl	String - 500	M	URL to receive webhook notifications	https://jul.adports.ae/api/webhooks
2	events	Array	M	Array of events to subscribe to	["status_changed", "amendment_approved"]
3	certificateIds	Array	O	Specific certificates to monitor	[503808, 503809]
4	authToken	String - 100	O	Authentication token for webhook	webhook_token_123

Webhook Payload Elements:

--	--	--	--	--	--

S. No	Attributes	Data Type-Length	Condition	Format/Derivation Logic	Data Example
1	eventType	String - 50	M	Type of event that occurred	status_changed
2	certificateId	Integer	M	Certificate that triggered event	503808
3	timestamp	DateTime	M	Event occurrence timestamp	2025-11-13T16:30:00Z
4	oldStatus	String - 20	O	Previous status (for status changes)	Submitted
5	newStatus	String - 20	O	New status (for status changes)	Approved
6	data	Object	O	Additional event-specific data	{...}

Sample Webhook Payload:

```
{
  "eventType": "status_changed",
  "certificateId": 503808,
  "timestamp": "2025-11-13T16:30:00Z",
  "oldStatus": "Submitted",
  "newStatus": "Approved",
  "data": {
    "approvedBy": "arcccla_broker_456",
    "approvalComments": "All documentation verified and approved",
    "invoiceNumber": "INV-503808-2025",
    "paymentDueDate": "2025-11-20T23:59:59Z"
  }
}
```

7. Validation Framework

7.1 Field-Level Validation Rules

Text Field Validations:

- **TXT_001:** Required fields cannot be null or empty
- **TXT_002:** Text fields must not exceed maximum length
- **TXT_003:** Special characters properly escaped for security

Numeric Field Validations:

- **NUM_001:** Numeric fields must contain valid numbers
- **NUM_002:** Currency amounts non-negative with max 2 decimal places
- **NUM_003:** Integer IDs must be positive

Date/Time Validations:

- **DT_001:** All dates in ISO 8601 format (YYYY-MM-DDTHH:mm:ssZ)
- **DT_002:** ETD must be before or equal to ETA
- **DT_003:** Creation dates cannot be in future

7.2 Business Logic Validation

Trade Compliance:

- **TC_001:** Angola must be origin or destination for CNCA certificates
- **TC_002:** Sanctioned countries require special compliance screening
- **TC_003:** High-value shipments (>\$50,000) require enhanced documentation

Financial Validations:

- **FIN_001:** Invoice value must match bill of lading value ($\pm 5\%$ tolerance)
- **FIN_002:** Currency conversion rates current (within 24 hours)
- **FIN_003:** Payment terms consistent with Incoterms selected

Workflow State Validations:

- **ST_001:** Only valid status transitions allowed per business rules
- **ST_002:** Required actions completed before status advancement
- **ST_003:** Role-based permissions validated for status changes

7.3 Error Handling Standards

Error Response Structure:

```
{
  "error": {
    "code": "CTN_V001",
    "message": "Bill of Lading number must be unique",
    "severity": "ERROR",
    "field": "BL_number",
    "timestamp": "2025-11-13T16:45:00Z"
  }
}
```

Performance Standards:

- **PERF_001:** Field validations complete within 100ms
- **PERF_002:** Business validations complete within 500ms
- **PERF_003:** External validations complete within 2 seconds

End of Interface Control Document

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